

TEXONO sensitivity to transverse hidden photons

A fresh calculation guided by Danilov et al.

Derivation, ROOT/C++ calculation and source audit

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1 Result and meaning of the limit

For the homogeneous-medium benchmark of Danilov et al.[1], the fresh calculation gives a one-sided Gaussian upper boundary $\epsilon_{95} = 1.33627 \times 10^{-5}$ at $M = 1$ keV, using transverse absorption, the original fiducial exposure and an assumed photon selection efficiency $\eta_A = 1/6$. This is a conditional reinterpretation of TEXONO, not a collaboration-calibrated exclusion. The full mass dependence is plotted below. The material resonance is omitted; a missing boundary at very low mass means that the crossing lies outside the small-mixing domain scanned here.

Danilov's source mechanism is coherent conversion of ordinary photons into the sterile component while the ordinary component is attenuated in the reactor. It preserves energy. It replaces the Compton-convolved source in this calculation; it is not multiplied by that source. Compton scattering still provides a detector interaction, and its production analogue is derived below as a separate microscopic check.

2 Model, inputs and scope of the derivation

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{4}X_{\mu\nu}X^{\mu\nu} - \frac{\epsilon}{2}F_{\mu\nu}X^{\mu\nu} + \frac{M^2}{2}X_\mu X^\mu - eA_\mu J_{\text{em}}^\mu. \quad (1)$$

There are no lighter hidden particles and hence no invisible decay channel. The tree-level scattering amplitudes, polarization traces, phase space, homogeneous conversion amplitude and event integral are derived and implemented anew. QED Feynman rules and standard phase-space identities are starting identities. The reactor photon spectrum and experimental information are measured or published inputs. The exact one-loop three-photon enhancement is a downloaded numerical input, not a new loop calculation. No previous project code, curves or numerical results are imported.

The reactor source is the spectrum quoted by Danilov from FRJ-1, valid above 0.2 MeV:

$$Q_\gamma(E) = 0.58 \times 10^{18} \frac{P_{\text{th}}}{\text{MW}} e^{-E/(0.91 \text{ MeV})} \quad [\text{s}^{-1}\text{MeV}^{-1}]. \quad (2)$$

This is an empirical source normalization, not a source derived from nuclear transition strengths. The original FRJ-1 annual report was located, but its server returned an HTML challenge instead of the PDF; the spectrum is therefore taken from Danilov. A reactor-specific spectral uncertainty is not available.

3 Production by conversion in an absorbing medium

Set $X = S - \epsilon A$ at leading order. For either transverse polarization the propagation equation is $i\partial_z \Psi = H\Psi$, with

$$H = \frac{1}{2E} \begin{pmatrix} m_\gamma^2 + \epsilon^2 M^2 - iE\Gamma & -\epsilon M^2 \\ -\epsilon M^2 & M^2 \end{pmatrix}, \quad m_\gamma^2 = \frac{4\pi\alpha n_e}{m_e}. \quad (3)$$

Here Γ^{-1} is the intensity attenuation length: an unmixed photon amplitude falls as $e^{-\Gamma z/2}$. This convention agrees with the denominator of Danilov Eq. (6) and Redondo Eqs. (2.5)–(2.10)[2]. Using $-i\Gamma$ in the Hamiltonian while calling Γ^{-1} the intensity length would introduce a factor-four error in the damping term. Natural units are used inside these equations.

$$\delta = \frac{m_\gamma^2 - M^2}{2E}, \quad g = \frac{\epsilon M^2}{2E}, \quad (4)$$

$$\mathcal{A}_{A \rightarrow S}(L) = g \int_0^L e^{i\delta z - \Gamma z/2} dz = g \frac{e^{(i\delta - \Gamma/2)L} - 1}{i\delta - \Gamma/2}, \quad (5)$$

$$P_{A \rightarrow S}(L) = \frac{\epsilon^2 M^4}{(M^2 - m_\gamma^2)^2 + E^2 \Gamma^2} [1 + e^{-\Gamma L} - 2e^{-\Gamma L/2} \cos(\delta L)], \quad (6)$$

$$P_R(E) \xrightarrow{\Gamma L \gg 1} \frac{\epsilon^2 M^4}{(M^2 - m_R^2)^2 + E^2 \Gamma_R^2} \equiv \epsilon^2 C_R(E, M), \quad Q_X(E) = Q_\gamma(E) P_R(E). \quad (7)$$

The last expression is the leading small-mixing form of Danilov's probability. The discriminant of the displayed real mass matrix is

$$(\Delta m^2)^2 = (M^2 - m_\gamma^2)^2 + 2\epsilon^2 M^2 (M^2 + m_\gamma^2) + \epsilon^4 M^4. \quad (8)$$

Danilov retains the term quadratic in mixing. The numerical calculation uses the leading denominator and requires $\epsilon M^2/|M^2 - m_{R,D}^2| < 0.1$. It also requires $|M^2 - m_{R,D}^2| > 10E_{\max} \Gamma_{R,D}$, so that the unmodeled narrow resonance is excluded and detector damping is subdominant. The benchmark has $m_R = m_D = 20$ eV and attenuation lengths of 10 cm; the detector length here only defines the resonance diagnostic. The paper's higher-density range is illustrated separately with 60 eV. These plasma masses are benchmark inputs, not a composition model of CsI.

There is no area-valued production cross section for this coherent one-particle conversion. The appropriate quantity is P_R per produced photon, or Q_X per unit energy and time. Replacing it by a fictitious two-body cross section would mix distinct descriptions. In the heavy limit $C_R \rightarrow 1$; at small mass $C_R \sim M^4/m_R^4$. No secondary reactor Compton cascade is added.

4 Compton production from the two QED diagrams

$$\gamma(k) + e(p) \rightarrow X(q) + e(p'), \quad p^2 = p'^2 = m^2, \quad k^2 = 0, \quad q^2 = M^2, \quad m = m_e, \quad (9)$$

$$\mathcal{M} = \epsilon e^2 \bar{u}(p') \mathcal{V}^{\nu\mu} u(p) \xi_\nu^* \varepsilon_\mu, \quad (10)$$

$$\mathcal{V}^{\nu\mu} = \gamma^\nu \frac{\not{p} + \not{k} + m}{s - m^2} \gamma^\mu + \gamma^\mu \frac{\not{p} - \not{q} + m}{u - m^2} \gamma^\nu. \quad (11)$$

Contracting either external momentum into the sum of diagrams and using $(\not{p} - m)u = 0$ cancels the two terms. Current conservation therefore removes the $q_\mu q_\nu / M^2$ term in the complete massive polarization sum. It does not mean that a physical longitudinal state is absent at finite mass. Expanding the spin trace and using the Clifford algebra gives

$$|\overline{\mathcal{M}}|^2 = \epsilon^2 e^4 F(a, b, c, d), \quad a = s - m^2, \quad b = u - m^2, \quad c = m^2, \quad d = M^2, \quad (12)$$

$$F = -2 \left(\frac{a}{b} + \frac{b}{a} \right) + 4(2c + d) \left[\frac{1}{a} + \frac{1}{b} + c \left(\frac{1}{a} + \frac{1}{b} \right)^2 - \frac{d}{ab} \right]. \quad (13)$$

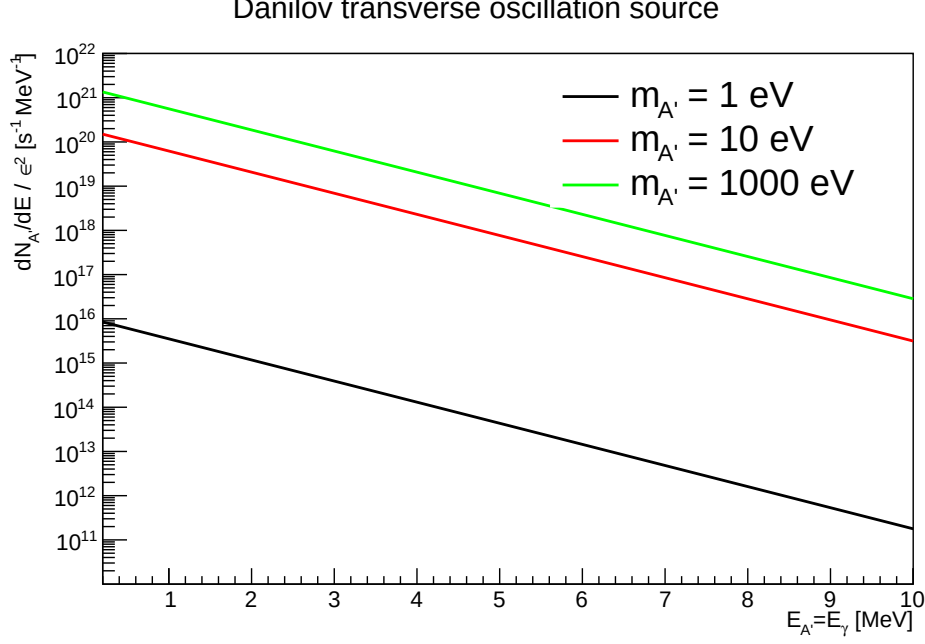


Figure 1: Newly calculated oscillation source, with mixing squared removed.

The initial average is over two electron spins and two photon polarizations. The derivation program contracts the gamma traces symbolically; an independent C++ calculation sums physical polarization vectors and checks the same expression.

$$E_{\gamma,\text{th}} = M + \frac{M^2}{2m}, \quad s = m^2 + 2mE_\gamma, \quad \lambda = [s - (m + M)^2][s - (m - M)^2], \quad (14)$$

$$E_X^\pm = \frac{(E_\gamma + m)(s + M^2 - m^2) \pm E_\gamma \sqrt{\lambda}}{2s}, \quad (15)$$

$$\frac{d\sigma_{\text{prod}}}{dE_X} = \frac{\epsilon^2 e^4 (\hbar c)^2}{32\pi m E_\gamma^2} F, \quad E_X^- \leq E_X \leq E_X^+. \quad (16)$$

Outside this support the cross section is zero. The Jacobian follows from $t = -2m(E_\gamma - E_X)$ and $d\sigma/dt = |\mathcal{M}|^2/[16\pi(s - m^2)^2]$. The implementation integrates in the center-of-mass angle instead:

$$\frac{d\sigma}{d\cos\theta_*} = \frac{(\hbar c)^2 p_f^*}{32\pi s p_i^*} |\mathcal{M}|^2. \quad (17)$$

5 Absorption with the produced polarization density matrix

An unpolarized ordinary-photon source gives $\rho_X = \text{diag}(1/2, 1/2, 0)$ in the helicity basis under the transverse conversion model. The incoming population must therefore be averaged over two transverse states, not three vector states. Longitudinal nuclear emission is a different source, suppressed by current conservation at high energy; it is not supplied by the transverse Hamiltonian. Its relative parametric size is M^2/E^2 , not a measured branching fraction.

For $X(q) + e(p) \rightarrow \gamma(k) + e(p')$, use the same two diagrams with the vector roles exchanged. Let $E = q^0$, $w = k^0$, $p_X = \sqrt{E^2 - M^2}$, and define

$$a = M^2 + 2mE, \quad b = -2mw, \quad c = m^2, \quad d = M^2, \quad s = m^2 + a. \quad (18)$$

The longitudinal trace, averaged over the electron spin but for a single incoming longitudinal vector, is derived independently as

$$D = (a - d)^2 - 4cd, \quad U = ab - 4ac + 2cd - 8c^2, \quad (19)$$

$$V = a^2b + a^2c + ab^2 - abd + 2abc + b^2c, \quad (20)$$

$$F_L = \frac{8dUV}{a^2b^2D}, \quad F_T = F - \frac{1}{2}F_L, \quad F_{\text{unpol}} = \frac{2F_T + F_L}{3} = \frac{2}{3}F. \quad (21)$$

This follows by inserting $\xi_L = (p_X, 0, 0, E)/M$ into the electron trace and summing the outgoing photon states. The code uses the Ward-equivalent vector $\xi_L - q/M$ to avoid cancellation of large E/M terms. The remaining transverse sum follows from completeness. SymPy verifies $F_L \rightarrow 0$ at $M \rightarrow 0$, and numerical physical vectors independently verify the full finite-mass expression.

$$w_{\pm} = \frac{(s - m^2)(E + m \pm p_X)}{2s}, \quad (22)$$

$$\frac{d\sigma_T^{\text{vac}}}{dw} = \frac{\epsilon^2 e^4 (\hbar c)^2}{32\pi m p_X^2} F_T, \quad \frac{d\sigma_L^{\text{vac}}}{dw} = \frac{\epsilon^2 e^4 (\hbar c)^2}{32\pi m p_X^2} F_L, \quad (23)$$

$$\sigma_T^{\text{vac}} \xrightarrow{M \rightarrow 0} \epsilon^2 \sigma_{\text{KN}}, \quad \sigma_{\text{unpol}}^{\text{vac}} \xrightarrow{M \rightarrow 0} \frac{2}{3} \epsilon^2 \sigma_{\text{KN}}. \quad (24)$$

For the propagating mostly hidden mode in detector matter, the photon admixture is $\epsilon M^2/(M^2 - m_D^2)$ away from resonance. Thus the bulk interaction cross section in the Danilov approximation is

$$\sigma_A^{\text{med}}(E, M) = \epsilon^2 C_D(M) \hat{\sigma}_T(E, M), \quad C_D = \frac{M^4}{(M^2 - m_D^2)^2}, \quad \hat{\sigma}_T = \sigma_T^{\text{vac}}/\epsilon^2. \quad (25)$$

Danilov Eq. (9) supplies the squared admixture. It is not the total probability for a count; the target electrons, cross section and selection must still be included. This is a bulk, weak-mixing approximation. Boundary transients and inhomogeneous material require a transport calculation.

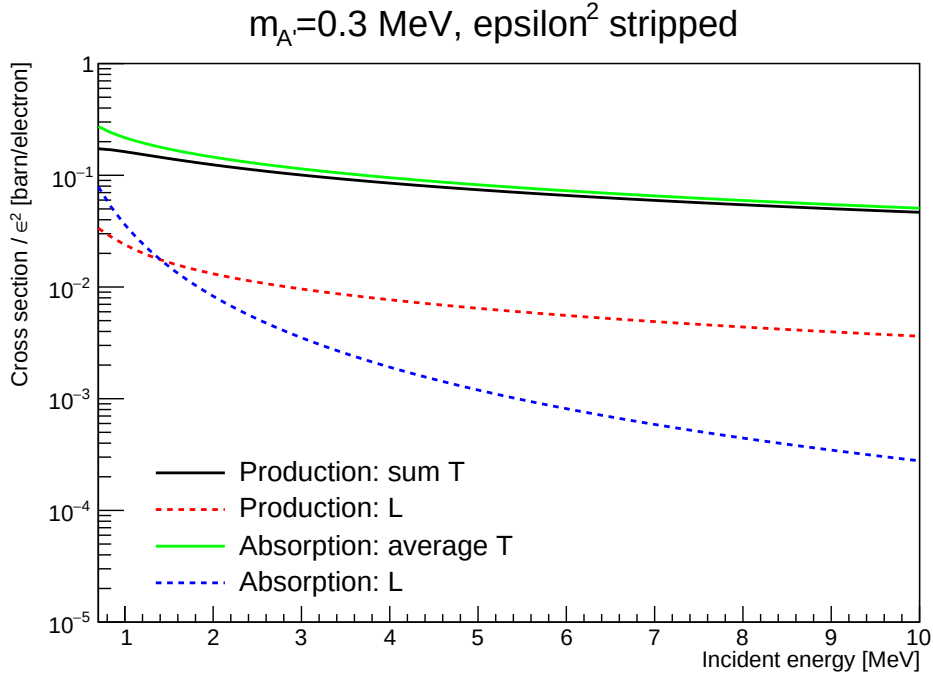


Figure 2: Fresh, polarization-resolved production and absorption. These are vacuum microscopic cross sections, not an extra reactor source.

6 Decay: survival and counts inside the detector

Izaguirre et al.[5] separate absorption after propagation from decay within the detector. Their electron-pair channel is closed for $M < 1 \text{ MeV} < 2m_e$. With no lighter dark particles, the leading allowed electromagnetic channel is $X \rightarrow 3\gamma$. We use the Euler–Heisenberg width multiplied by the published full-loop enhancement of McDermott et al.[6]:

$$\Gamma_{3\gamma} = \frac{17\epsilon^2\alpha^4}{11664000\pi^3} \frac{M^9}{m_e^8} f(M), \quad (26)$$

$$\ell(E) = \frac{\hbar c \sqrt{E^2 - M^2}}{M\Gamma_{3\gamma}}, \quad S(E) = e^{-R/\ell(E)}, \quad (27)$$

$$P_{\text{decay in chord}} = S(E)[1 - e^{-L/\ell(E)}]. \quad (28)$$

The tabulated $f(M)$ is interpolated logarithmically; below its first mass point the published low-mass series is used. No extrapolation beyond its upper endpoint is allowed. Rotational invariance makes the total rest-frame decay width independent of initial spin orientation; polarization still affects angular acceptance, which is not inferred from that width. At the absorption-only boundary, the largest propagation exponent is 1.08389×10^{-8} . This makes loss over the baseline negligible. It does not prove that detector decays are negligible compared with absorption. In the thin-signal and point-source limit, integrating detector chords gives $\int dA L = V$, so both terms can be included without choosing an arbitrary chord:

$$N = \frac{T}{4\pi R^2} \int_{E_{\min}}^{E_{\max}} dE Q_\gamma(E) P_R(E) S(E) \left[\eta_A N_e \sigma_A^{\text{med}}(E) + \eta_D \frac{V}{\ell(E)} \right]. \quad (29)$$

Here $N_e = M_{\text{det}}[\text{kg}] 10^3 N_A Z/A$, $V = M_{\text{det}}[\text{kg}] 10^3 / \rho$, and the CsI density is 4.51 g cm^{-3} from NIST. Full energy containment gives $E_{\text{vis}} = E$. The main bound sets $\eta_D = 0$ because TEXONO has not published the three-photon signal acceptance. A second curve shows $\eta_D = \eta_A$ as an explicit response hypothesis. It strengthens the upper-mass sensitivity and must not be presented as measured efficiency.

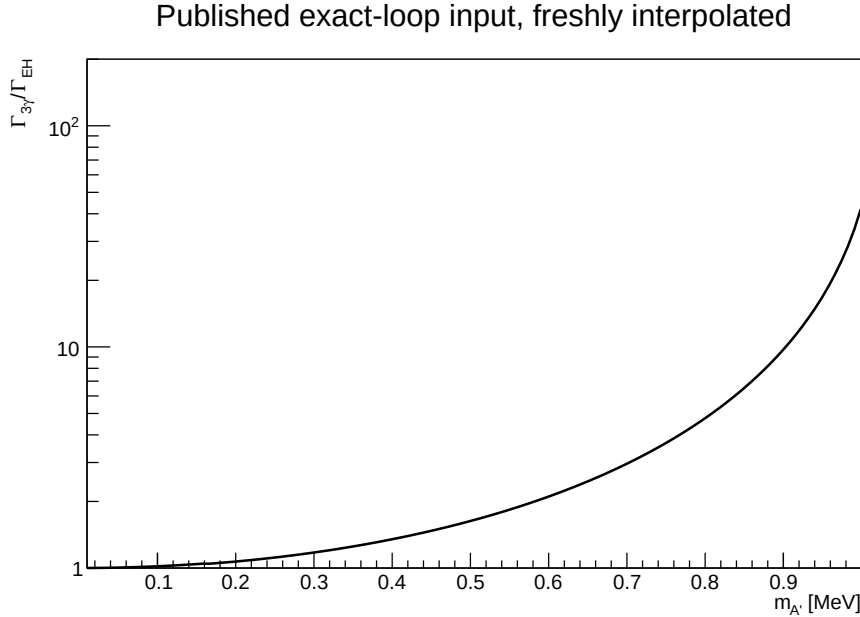


Figure 3: Published loop enhancement, freshly interpolated. The loop amplitude is not rederived here.

7 TEXONO inputs, statistics and the exclusion plane

The original TEXONO paper[4] gives a fiducial reactor-ON exposure of 29882 kg day and a 3–8 MeV window. The code represents this as 187 kg times the corresponding effective live time. The baseline is 28 m and thermal power 2.9 GW. CsI has $Z = 55 + 53 = 108$ electrons per formula unit and $A = 259.8099219 \text{ g mol}^{-1}$; the tiny Tl doping is neglected. Danilov quotes an excess 30.7 ± 100.6 events. The one-sided Gaussian construction used here is

$$N_{95} = 30.7 + \Phi^{-1}(0.95)100.6 = 196.17227, \quad N(M, \epsilon_{95}) = N_{95}. \quad (30)$$

Using 1.64 rather than the full Gaussian quantile gives Danilov’s rounded 195.7 events. Their 1174.1 preselection cap comes from the assumed anti-Compton suppression of about six. We keep the selected-event cap and put $1/6$ in η_A , avoiding a second application of the same correction. TEXONO Table III measures background survival 0.16 under the multi-hit veto; it does not calibrate hidden-photon signal acceptance. The Gaussian construction has no profiling of a new detector response or spectral nuisance model. Away from resonance and with negligible baseline decay, $N = K(M)\epsilon^4$. Therefore $\epsilon_{95} = (N_{95}/K)^{1/4}$. At low mass, the two medium factors yield $K \propto M^8$ and $\epsilon_{95} \propto M^{-2}$. The C++ solver includes survival and solves the full integral; the scaling is tested independently. Larger mixings are excluded only within the stated weak-mixing and transparent-signal regime.

M [eV]	$\epsilon_{95} (\eta_D = 0)$	$\epsilon_{95} (\eta_D = \eta_A)$
1	5.33434×10^{-3}	5.33434×10^{-3}
10	4.01106×10^{-5}	4.01106×10^{-5}
1000	1.33627×10^{-5}	1.33627×10^{-5}
100000	1.33712×10^{-5}	1.33712×10^{-5}
999000	1.34482×10^{-5}	1.29767×10^{-5}

8 Comparison, validity and what is not established

A direct rescaling of Park’s printed 2.1×10^{-5} bound[7], by the event-cap change and Danilov’s transverse $3/2$ rate correction, gives 2.96412×10^{-5} . The independent source integral gives 1.33627×10^{-5} on the plateau. This comparison fails a ten-percent reproduction criterion. It is not repaired by fitting the source normalization. Danilov describes a recalculation of Park; the stated energy-preserving source and the earlier Compton convolution are different prescriptions. The result here is a fresh implementation of the explicit source probability and photon spectrum, not a numerical reproduction of the published curve.

Exact finite-mass two-body kinematics is used for scattering throughout $0 < M < 1 \text{ MeV}$. The coherent production Hamiltonian assumes $M/E \ll 1$. At the endpoint and the bottom of the ROI, $(M/E)^2$ can reach about 0.11. This is an expansion parameter, not a calibrated error bar: nuclear transition multipoles and longitudinal emission have not been reconstructed. Below 0.1 MeV the same parameter is at most about 0.0011. Atomic binding is negligible relative to the MeV energies for the free-electron approximation, but material attenuation, photon escape and selection need detector transport. The plotted 20 eV benchmark is retained to follow Danilov; actual CsI and reactor profiles can change the low-mass result.

Checked: symbolic total and longitudinal traces; both Ward identities; massless Klein–Nishina limits; massive detailed balance; doubled angular and energy quadrature; the rounded Danilov count cap; published decay-table knots; and closure of every reported event boundary. Not established: the full TEXONO likelihood, energy-dependent acceptance for either channel, material profiles and resonances, secondary reactor transport, longitudinal nuclear production and the exact massive

TEXONO: one-sided Gaussian 95% CL, conditional response

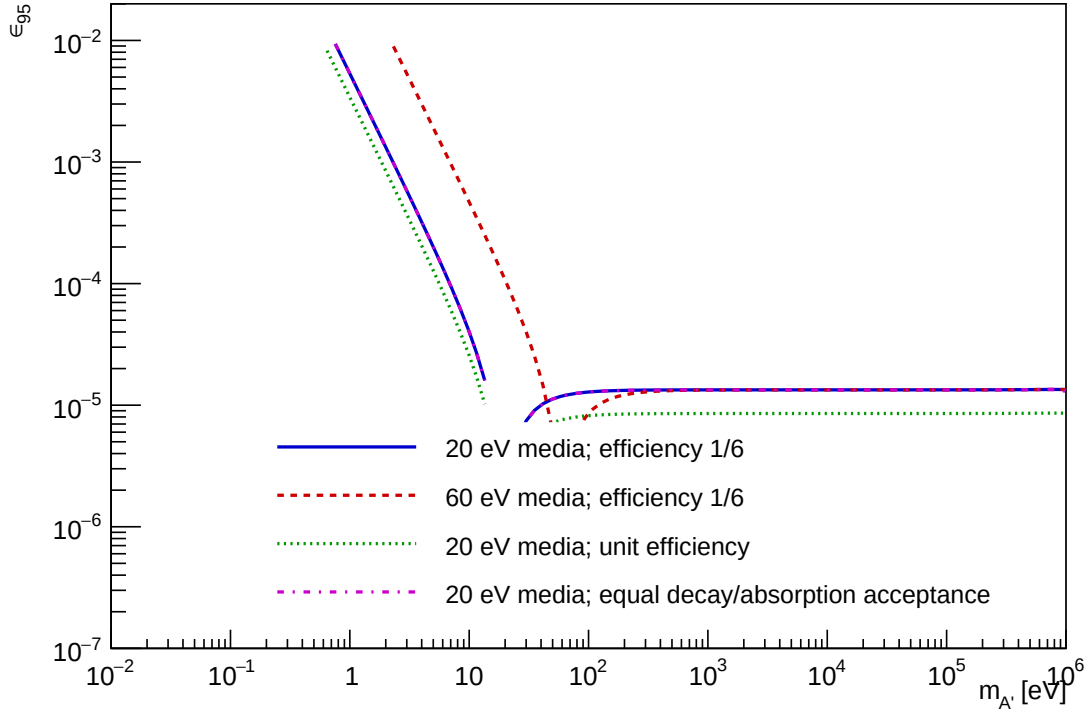


Figure 4: Fresh conditional boundaries. Curves split across excluded resonance intervals. The mass scan stops below 1 MeV. The upper-mass source still uses a relativistic approximation.

nuclear-source correction. These limitations prevent presenting the curve as a detector-validated experimental exclusion.

9 Reusing the calculation

Run `bash work/danilov_fresh/run.sh`. Edit `experiment.conf`: power, distance, exposure, mass, Z/A , density, ROI, separate absorption/decay efficiencies, medium parameters and the event cap. A negative cap requests the Gaussian excess/error construction; otherwise an externally justified positive cap is used. For a different background likelihood, supply its limit rather than treating a raw count as Gaussian. The present sub-MeV scan requires $E_{\min} > 1$ MeV and $\epsilon \leq 0.01$. Lower-energy detectors require atomic response and a revised source model. All physical calculations are named C++ functions; SymPy supplies independent algebra and marimo displays the freshly generated ROOT figures.

References

- [1] M. Danilov, S. Demidov, D. Gorbunov, arXiv:1804.10777v2, PRL 122, 041801 (2019), Eqs. (2), (5), (6), (9), TEXONO discussion.
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- [3] J. Redondo, G. Raffelt, arXiv:1305.2920; H. An, M. Pospelov, J. Pradler, arXiv:1302.3884. Polarization and in-medium mixing; corpus PDFs retained as sources.

- [4] M. Deniz et al., arXiv:0911.1597, PRD 81, 072001 (2010), Tables II–III and Sec. VII. New corpus addition.
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- [7] H. Park, arXiv:1705.02470, PRL 119, 081801 (2017); printed TEXONO bound used only for comparison.

All cited PDFs and the downloaded decay table are in `papers/danilov_fresh/`. NIST CsI composition/density is archived there as `nist_csi.html`.